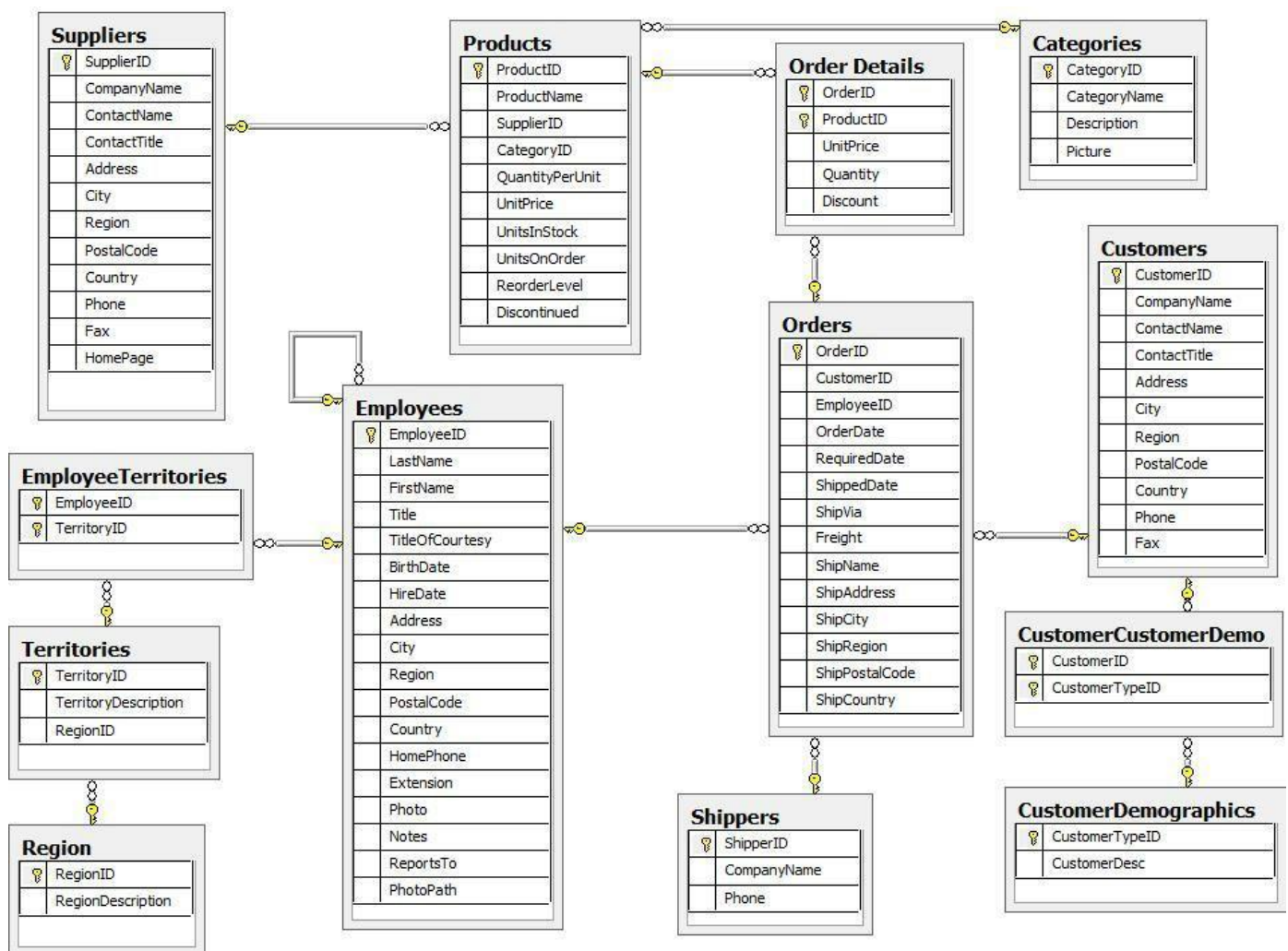


The Northwind Database

The Northwind database is a sample database used by Microsoft to demonstrate the features of some of its products, including SQL Server and Microsoft Access. The database contains the sales data for Northwind Traders, a fictitious specialty foods export-import company.

Although the code taught in this class is not specific to Microsoft products, we use the Northwind database for many of our examples because many people are already familiar with it and because there are many resources for related learning that make use of the same database.

The diagram below shows the table structure of the Northwind database.



Exercises

1. Select the FirstName and LastName columns from the Employees table. Sort by LastName.
2. Create a report showing Northwind's orders sorted by Freight from most expensive to cheapest. Show OrderID, OrderDate, ShippedDate, CustomerID and Freight.
3. Create a report showing the title and the first and last name of all sales representatives.
4. Create a report showing the first and last names of all employees who have a region specified.
5. Create a report showing the first and last name of all employees whose last names start with a letter in the last half of the alphabet. Sort by LastName in descending order.
6. Create a report showing the title of courtesy and the first and last name of all employees whose title of courtesy begins with "M".
7. Create a report showing the first and last name of all sales representatives who are from Seattle or Redmond.
8. Create a report that shows the company name, contact title, city and country of all customers in Mexico or in any city in Spain except Madrid.
9. If the cost of freight is greater than or equal to \$500.00, it will now be taxed by 10%. Create a report that shows the order id and freight cost with this tax for all orders of \$500 or more.
10. Find the Total Number of Units Ordered of Product ID 3
11. Retrieve the number of employees in each city
12. Retrieve the number of suppliers in each country
13. Find the number of sales representatives in each city that contains at least 2 sales representatives. Order by the number of employees.
14. Find the Companies (CompanyName) that placed orders in 1997
15. Create a report showing employee orders.
16. Create a report showing the Order ID, the name of the company that placed the order and the first and last name of the associated employee. Only show orders placed after January 1, 1998 that shipped after they were required. Sort by Company Name.
17. Create a report that shows, by product name, the total quantity of products (from the Order_Details table) ordered by product. Only show the 5 records corresponding to the products which were less ordered for which the quantity ordered is fewer than 650.

- 18.** Create a report showing all distinct countries stored in database
- 19.** Change the database so that the table Products can be a lookup table
- 20.** Create a procedure for inserting a new customer
- 21.** Create a procedure for updating data on an existing customer
- 22.** Create a procedure for deleting all products never ordered and keep the information on a historic table
- 23.** Create a procedure for inserting a new product
- 24.** Create a procedure for updating data on an existing product
- 25.** Change the database so that information about countries is stored in a lookup table with three fields: Country Id, Country Name and Country Code. Modify all the previous queries accordingly.